

College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Midterm Examination of Semester 1

Year: 2017

Subject: 353152 Electronic Circuits I

Section: 05-06

Date: 2 October 2017

Time: 8.00-10.00

Name: _____ ID: _____ Field of Study: _____

Instructions:

1. The examination has 13 pages (including this page), 2 sections (37 questions) and a total score of 89 points.
2. Write all your solutions and answers on this examination sheet.
3. This is a closed book examination.
4. You are not allowed to leave the exam room during the first 1 hour after the beginning of the exam.
5. You are not allowed to open the exam papers or start to answer before the proctor's permission.
6. You are not allowed to use the restroom during the exam except in case of an emergency.
7. No documents are allowed to be taken out of the examination room.
8. Calculators is allowed in the examination.
9. Electronic communication devices are NOT allowed in the examination room.
10. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

Part 1 (64 points) Multiple choices

Instruction: Mark the correct answer for the following questions in the provided answer sheet.

-
1. Valence shell contain up to _____ electrons.
(a) 2 (b) 4 (c) 6 (d) 8

 2. Which type of material has widest energy gap between conduction band and valence band?
(a) insulator (b) semiconductor
(c) conductor (d) resistor

 3. Atoms of semiconductors contain _____ valence electrons.
(a) 2 (b) 4 (c) 6 (d) 8

 4. How much voltage drop over silicon diode when it is in forward bias mode?
(a) 0.2 V (b) 0.3 V (c) 0.6 V (d) 0.7 V

 5. Which one of the following is true?
(a) Diode is open circuit when operating in forward bias mode.
(b) Forward bias means the voltage on p junction is lower than n junction.
(c) For ideal diode, there is no voltage drop for forward bias mode.
(d) The voltage drops of a Silicon diode and Germanium diode are at 0.7 V when they are operating in forward bias mode.

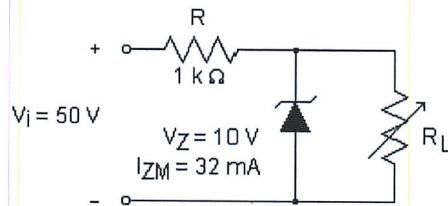
 6. Which circuit provides output as $V_{DC} = 0.318V_m$ for ideal case?
(a) Half Wave Rectifier (b) Bridge Rectifier
(c) Full-Wave Rectifier (d) Center-Tapped Transformer Rectifier

7. Which one of the following is true?
- (a) Bridge-rectifier has lower voltage drop at diode compare with center-tapped transformer rectifier.
 - (b) The cost of bridge-rectifier is higher than center-tapped transformer rectifier.
 - (c) Half-wave rectifier provides higher dc voltage output than full-wave rectifier.
 - (d) Bridge-rectifier contains more number of diodes over center-tapped transformer rectifier.
8. Which of the following transistor operating mode represents the short circuit?
- (a) Cut-off
 - (b) Saturation
 - (c) Active
 - (d) None
9. Which of the following is true?
- (a) EB Junction is forward biased when transistor is in active mode.
 - (b) EB Junction is forward biased when transistor is in cutoff mode.
 - (c) CB Junction is reverse biased when transistor is in saturation mode.
 - (d) CB Junction is forward biased when transistor is in cutoff mode.
10. Which diode(s) has (have) a zero level current and voltage drop in the ideal model?
- (a) Si
 - (b) Ge
 - (c) Both Si and Ge
 - (d) Neither Si nor Ge
11. An open circuit can have any voltage across its terminals, but the current is always ____.
- (a) 5 A
 - (b) 0 A
 - (c) 1 A
 - (d) ∞
12. The intersection of the load line with the characteristic curve determines the ____ of the system.
- (a) point of operation
 - (b) load-line analysis
 - (c) characteristic curve
 - (d) forward bias
13. The quiescent point (Q-point) is defined by a(n) ____.
- (a) ac network
 - (b) dc network
 - (c) ac and dc network
 - (d) KVL

14. A germanium diode is approximated by _____ equivalent for voltages less than 0.3 V.

- (a) a short circuit (b) series circuit
(c) a parallel circuit (d) an open circuit

15. With this Zener diode in its "on state," what is the level of I_Z (current of Zener diode) for the maximum load resistance?

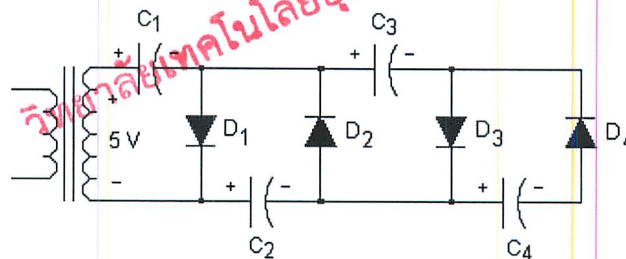


- (a) 0 mA (b) $I_Z = 8\text{mA}$ (c) $I_Z = I_{RL} = 40\text{mA}$ (d) $I_Z = 32\text{mA}$

16. What is the peak inverse voltage across each diode in a voltage doubler?

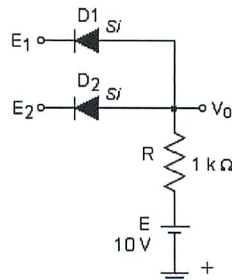
- (a) V_m (b) $2V_m$ (c) $0.5V_m$ (d) $0.25V_m$

17. What is the voltage measured from the negative terminal of C_4 to the negative terminal of the transformer?



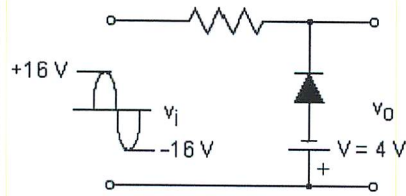
- (a) -10 V (b) -20 V (c) 10 V (d) 20 V

18. Determine the current through each diode if $E_1 = E_2 = 0\text{ V}$.



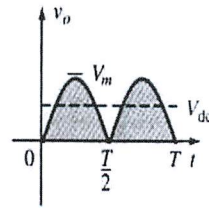
- (a) 4.65 mA (b) 9.3 mA (c) 18.68 mA (d) 0.7 mA

19. Determine the peak for both half cycles of the output waveform. Assume that the diode is ideal.



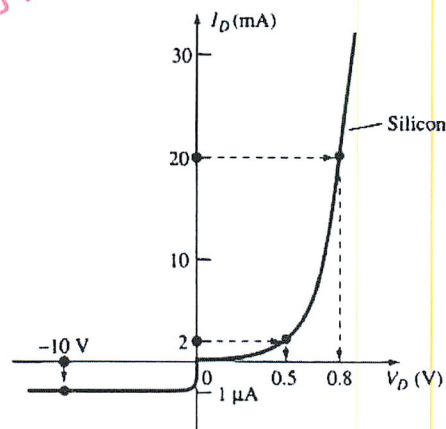
- (a) 16 V, -4 V (b) 16 V, 4 V (c) -16 V, 4 V (d) -16 V, -4 V

20. Which of the following is the average output value of full-wave rectifier?



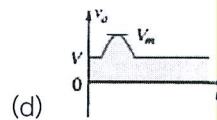
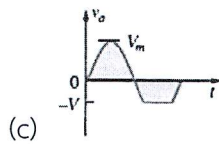
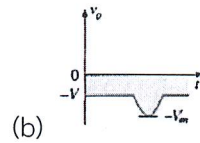
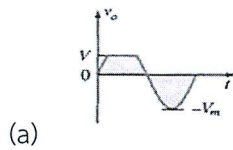
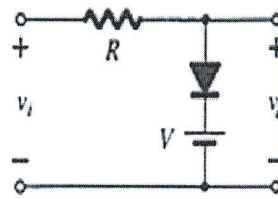
- (a) $0.318V_m$ (b) $0.5V_m$ (c) $0.636V_m$ (d) $1.414V_m$

21. From figure below, what is the DC resistance at 20 mA?

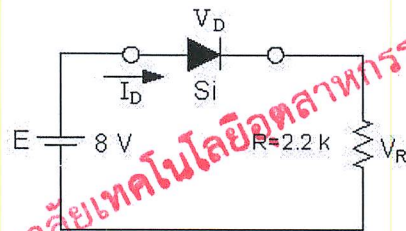


- (a) $4 \text{ m}\Omega$ (b) 4Ω (c) 40Ω (d) $40 \text{ k}\Omega$

22. What is the output voltage of circuit below? Assume that the diodes are ideal.



23. Find V_D and V_R



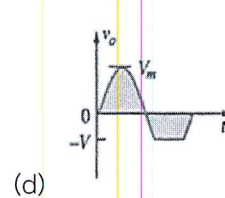
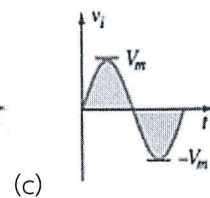
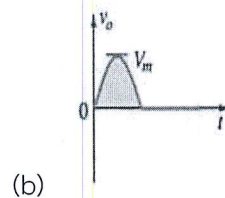
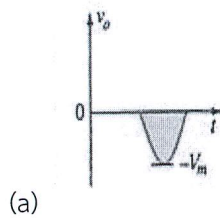
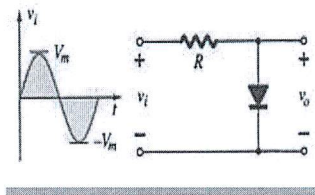
(a) 0.7V and 7.3V

(b) 0.3V and 7.7V

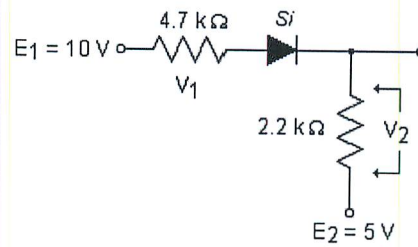
(c) 0.7V and 0.3V

(d) 0.3 V and 8V

24. What is the output voltage of the ideal diode circuit as shown in figure below?

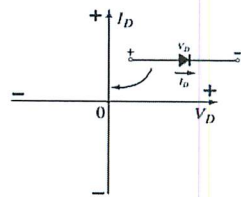
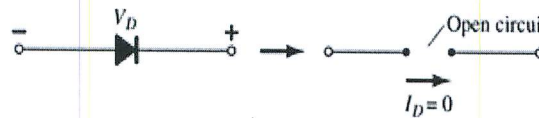


25. Determine V_2 .

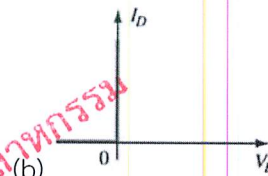


- (a) 3.201 V (b) 0 V (c) 4.3 V (d) 1.371 V

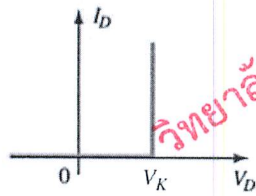
26. The ideal diode acts as an open switch. Which one is this diode characteristic?



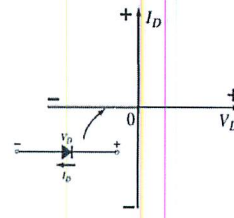
(a)



(b)

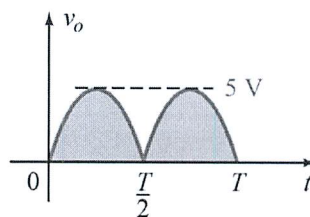


(c)



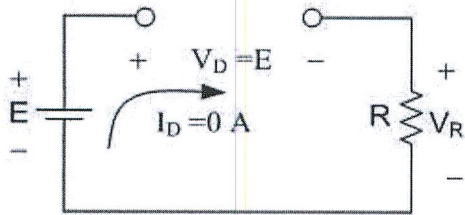
(d)

27. Which of the following is the V_{dc} of the waveform obtained from the silicon bridge rectifier?



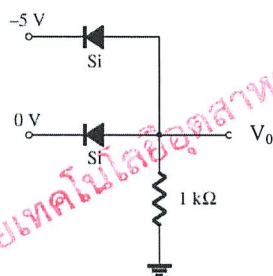
- (a) 3.18 V (b) 5 V (c) 2.29 V (d) 1.59 V

28. Find V_R where the given $E=20V$.



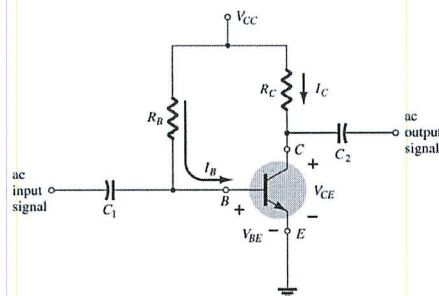
- (a) 0.7V
- (b) 0.3V
- (c) 0 V
- (d) 20V

29. Determine V_0 from ideal diode circuit below.



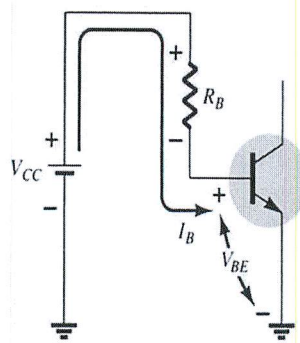
- (a) -5 V
- (b) -4.3 V
- (c) 0.7 V
- (d) 1.4 V

30. What is a bias type of the circuit in figure below?



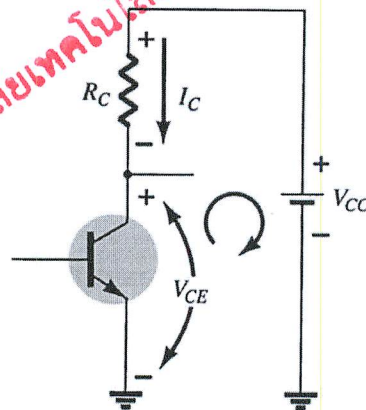
- (a) Fixed-bias circuit
- (b) Emitter-stabilized bias circuit
- (c) Voltage divider bias circuit
- (d) DC bias with voltage feedback

31. From Base-Emitter loop in picture below, what is the Kirchhoff's voltage equation for the loop?



- (a) $V_{CC} - I_B R_B - V_{BE} = 0$ (b) $V_{CC} - I_B R_B + V_{BE} = 0$
 (c) $V_{CC} + I_B R_B - V_{BE} = 0$ (d) $V_{CC} + I_B R_B + V_{BE} = 10$

32. From Collector-Emitter loop in picture below, what is the Kirchhoff's voltage equation for the loop?

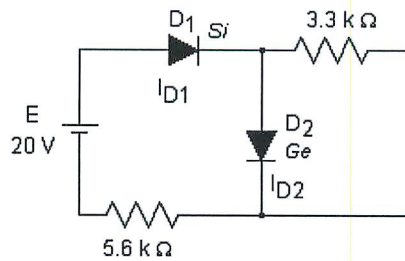


- (a) $V_{CE} + I_C R_C + V_{CC} = 0$ (b) $V_{CE} - I_C R_C + V_{CC} = 0$
 (c) $V_{CC} - I_B R_B - V_{BE} = 0$ (d) $V_{CE} + I_C R_C - V_{CC} = 0$

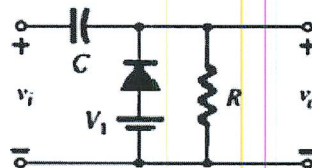
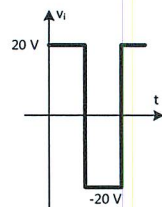
Part 2 (25 points) Calculation

Instruction: Show the mathematical expression and answers of following problems.

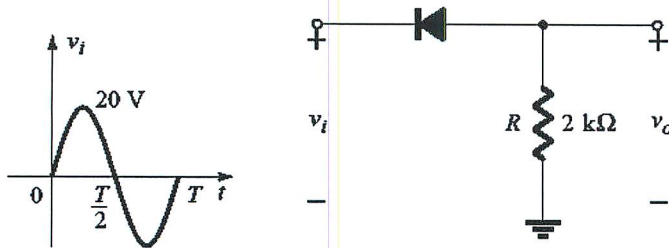
1. Determine I_{D1} from figure below



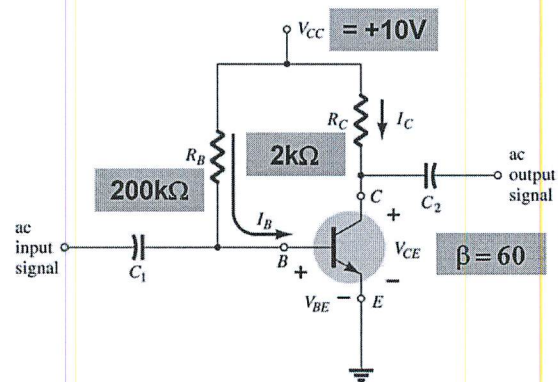
2. Sketch the output waveform of the network below. Given $V_1=5V$. Assume that the diode is ideal.



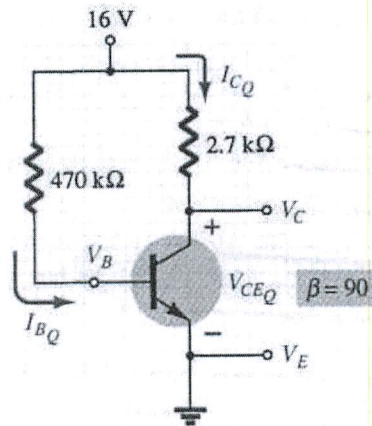
3. From the circuit, sketch the output and determine dc output voltage. Assume that the diode is ideal.



4. Determine I_B and I_C of the circuit below.



5. Determine $I_{C_{sat}}$ and $V_{CE_{cutoff}}$ of the circuit below.



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Answer Sheet

Name: _____ ID: _____

Subject: 353152 Electronic Circuits I

	a	b	c	d
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